

Beyond the Delayed-Feedback Ceiling: Offline Teachers as a Scaling Path for Conversion Prediction

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Abstract. Conversion labels arrive days after the click, so a model trained today must either wait for labels and train on stale data, or use fresh data whose labels are only partly observed. The standard answer is to correct the censoring. We argue the answer should instead be to distil an offline teacher, and we give both the theory and the measurements for why. Two ceilings govern the choice. Censoring correction is bounded by the information remaining in censored labels, a property of the delay distribution and therefore fixed; we derive that bound in closed form from the empirical delay distribution alone, before any model is trained, and on the Criteo delayed-feedback corpus it is 0.868 of the penalty for a seven-day window. Distillation is bounded instead by the teacher, which is a training-time asset and can be bought. We test both with 1,600 runs spanning five student capacities, twenty rolling origins and five strategies under one protocol. Correction attains 0.883, 0.874 and 0.849 at the three smallest capacities, within 0.02 of the prediction, confirming the theory and leaving little left to win. Distillation is not bounded that way: at a one-day window it beats the infeasible oracle at four of five capacities. Its gain tracks teacher headroom and collapses to -0.00015 once that headroom falls to 0.0036, making offline teacher scale the one lever here whose supply is not fixed by the corpus.

Keywords: Delayed feedback · Conversion rate prediction · Knowledge distillation · Censored labels · Computational advertising

1 Introduction

Conversion feedback is late. A click can convert minutes or weeks later, so at the moment a training set is assembled some of its labels are simply not yet determined. A decade of work treats this as an estimation problem to be solved on the label: wait for labels and accept stale covariates, or use fresh covariates and correct the resulting censoring, typically through a joint model of conversion and delay [1] or through importance weighting [4, 8, 6].

Meanwhile the systems these models live in have changed shape. Industrial advertising stacks increasingly concentrate model capacity in a single large, centrally trained model, of which Meta’s GEM [5] is a recent public example, while

the models that actually serve traffic remain small enough to meet a latency budget. That separation matters here for a specific reason. Training-time capacity and serving-time capacity have become different budgets, and only the second is bounded by an SLA. A strategy that converts training-time capacity into serving-time accuracy is therefore buying with the cheaper currency. Knowledge distillation is exactly such a strategy, and it has an obvious trajectory: as offline teachers grow towards LLM scale, whatever distillation buys grows with them.

This paper asks whether that trajectory is worth taking for delayed feedback, and answers it by separating two ceilings that the literature tends to conflate.

The first is a property of the data. A censoring correction can only use the information that survives in a partly observed label, and that information is fixed by the delay distribution. We make this quantitative: under the standard conditional-independence assumption, the recoverable fraction of the delay penalty is the mean relative Fisher information of the censored rows (Proposition 1), computable from the empirical delay distribution and the calendar before any model is trained. On the Criteo delayed-feedback corpus it is 0.868 for a seven-day window, and measured recovery is 0.883, 0.874 and 0.849 at our three smallest capacities, within 0.02 of a formula fitted to nothing. This is a good result for the theory and a discouraging one for the practice: correction is already at its ceiling, and that ceiling cannot be raised, because the delay distribution is not ours to change.

The second ceiling is a property of the budget. Distillation is not bounded by the oracle at all (Proposition 2): every other strategy here fits the student class to some coarsening of the oracle’s own data and therefore cannot beat it, whereas distillation routes its target through a strictly richer teacher class. That is not a technicality. At $\Delta = 1$ d, distillation beats the infeasible oracle at four of five capacities, by +0.0015 to +0.0040 normalised entropy, $p < 0.001$. No correction-based strategy exceeds the oracle anywhere at $\Delta = 7$ d.

We then show the gain is bought with exactly the quantity the LLM-teacher trajectory supplies. Defining teacher headroom as the teacher’s advantage over the waiting baseline, distillation’s gain is roughly flat across a $2.8\times$ range of headroom and then collapses to -0.00015 at the one cell where headroom falls to $+0.0036$. Distillation helps precisely while the teacher is meaningfully better than the student, and our teacher, at $2,262\times$ the student’s inference cost, has already run out of headroom against our largest student. Scaling the teacher is thus not an optional refinement of this result; it is the only direction in which the result can continue.

This study is a controlled baseline, not a demonstration with an LLM-scale teacher, which we did not run. Its job is to establish that the mechanism works, to measure what governs it, and to show that the alternative is already exhausted. Concretely: (i) a parameter-free ceiling on what censoring correction can recover, validated to within 0.02 (Sects. 3, 5.2); (ii) the argument and the evidence that distillation is not subject to it, including where it exceeds the infeasible oracle (Sect. 5.3); (iii) a measurement of what distillation’s advantage is bought with,

Table 1. The five strategies, with the infeasible Oracle and the distillation control. All are evaluated on the same held-out day against the same target (1), and all serve the same student

	training set	target
Oracle	$\mathcal{M} \cup \mathcal{C}$	true y (infeasible at T)
Wait	$\mathcal{M}$	y
Distil	$\mathcal{M} \cup \mathcal{C}$	teacher’s $\hat{\eta}$; teacher trained on $\mathcal{M}$
Distil-Stale	$\mathcal{M}$	the same teacher’s $\hat{\eta}$ (control)
Correct	$\mathcal{M} \cup \mathcal{C}$	y_{obs} ; loss on $\sigma(z)\hat{F}(e)$, serve $\sigma(z)$
Band	$[0, T - u_k), k = 1 \dots K$	$\mathbf{1}[u_{k-1} < d \leq u_k]$, predictions summed
Reweight	$\mathcal{M}$	y , weighted by $\hat{p}_{\mathcal{C}}(x)/\hat{p}_{\mathcal{M}}(x)$

using a control separating capacity transfer from delay mitigation that we argue should be mandatory here (Sects. 5.4, 5.5).

2 Problem Setting

A click occurs at time t with covariates x . If it converts, the conversion follows after a delay $d > 0$; write $d = \infty$ for a click that never converts. Under an attribution window Δ the prediction target is

$$y = \mathbf{1}[d \leq \Delta], \quad \eta(x) = \Pr(d \leq \Delta \mid x). \quad (1)$$

The difficulty is not that y is noisy but that it is not yet determined. At an origin T , the moment a training set is assembled, a row clicked at t has been observable for

$$e(t) = \text{clip}(T - t, 0, \Delta), \quad y_{\text{obs}} = \mathbf{1}[d \leq e(t)] \leq y. \quad (2)$$

Two regimes follow. If $t \leq T - \Delta$ then $e = \Delta$ and $y_{\text{obs}} = y$ exactly. If $t > T - \Delta$ the row is right-censored: $y_{\text{obs}} = 1$ is conclusive, but $y_{\text{obs}} = 0$ is ambiguous, because the click may still convert inside its remaining window. Everything below follows from (2), and no strategy is granted an assumption the others lack.

2.1 Five Strategies as Estimators

Let $\mathcal{M} = [0, T - \Delta)$ denote the mature rows and $\mathcal{C} = [T - \Delta, T)$ the censored ones, with $|\mathcal{M}| = m$ and $|\mathcal{C}| = c$. Table 1 states each strategy as a choice of training set and target. Every strategy serves the same student architecture, so they are indistinguishable in serving cost and differ only in what they train on. The Oracle is infeasible at time T , since it uses labels that do not yet exist; it serves only as the denominator for how much of the penalty a feasible strategy recovers.

Correct is the classical joint conversion-delay likelihood of [1], with a non-parametric $\hat{F}$. Under the conditional-independence assumption

$$\Pr(d \leq e \mid x, d \leq \Delta) = F(e), \quad (3)$$

that is, delay independent of the covariates given conversion inside the window, the observed label has mean $\Pr(y_{\text{obs}} = 1 \mid x, e) = \eta(x)F(e)$. Fitting $\sigma(z(x))\hat{F}(e)$ against y_{obs} under log loss is then a proper composite likelihood for η . The correction lives in the loss, and $\sigma(z)$ is served directly with no post-hoc rescaling.

Distil-Stale is not an ablation but a necessary control. Distil changes two things at once relative to Wait: the student inherits a teacher that may be larger than itself, and it trains on covariates that Wait had to discard. Distil-Stale removes only the second. Without it, any gain from Distil is unattributable, and Sect. 5.5 shows the attribution is not the one a reader would guess.

3 Two Ceilings

3.1 What Correction Can Recover

Under (3) a censored observation with window e is Bernoulli with mean $\eta F(e)$. Its Fisher information about η is $I(\eta; e) = (\partial_\eta \eta F)^2 / (\eta F(1 - \eta F)) = F / (\eta(1 - \eta F))$, against $I_{\text{full}} = 1 / (\eta(1 - \eta))$ for a mature row. The relative information of a censored row is therefore

$$\rho(F) = \frac{I(\eta; e)}{I_{\text{full}}} = \frac{F(1 - \eta)}{1 - \eta F} \in [0, 1], \quad \rho(1) = 1, \quad \rho(F) \approx F \text{ for small } \eta. \quad (4)$$

Proposition 1. *Assume (3), and that excess risk scales as $1/n$ in the effective sample size. Let $\bar{\rho} = |\mathcal{C}|^{-1} \sum_{i \in \mathcal{C}} \rho(F(e_i))$. Then the fraction of the Oracle-minus-Wait gap recovered by Correct is*

$$R = \frac{1/m - 1/(m + \bar{\rho}c)}{1/m - 1/(m + c)} = \frac{\bar{\rho}(m + c)}{m + \bar{\rho}c}. \quad (5)$$

R contains no fitted quantity: $\bar{\rho}$ follows from the empirical delay distribution and the base rate, and m and c from the calendar. It can be computed before training anything.

On this corpus the delay distribution is strongly front-loaded, with 57.4% of conversions within seven days landing in the first hour, so censored rows retain most of their information even when their window has barely opened. Averaged over the twenty origins used below, $\bar{\rho} = 0.886$ and $R = 0.888$ for $\Delta = 1$ d, and $\bar{\rho} = 0.847$ and $R = 0.868$ for $\Delta = 7$ d.

3.2 Why Distillation Is Not Subject to It

Proposition 1 prices one strategy, but the Oracle bounds a whole family.

Proposition 2. *Let $\mathcal{F}$ be the student class. Consider any strategy that fits $\mathcal{F}$ to a target measurable with respect to the Oracle’s own training data. Asymptotically its risk is at least the Oracle’s, so its recovery satisfies $R \leq 1$. Correct, Band and Reweight are of this form. Distil is not: its target is produced by a fit in a strictly richer class $\mathcal{G} \supset \mathcal{F}$. As the teacher approaches η , Distil’s limiting risk is the approximation error of $\mathcal{F}$ alone, whereas the Oracle additionally carries an estimation term of order $1/(m+c)$. Hence $R > 1$ is attainable for Distil and not for the others.*

The first half is a data-processing argument: y_{obs} , the band indicators and Reweight’s weights are all functions of (x, y, d) observed on a subset of the Oracle’s rows, so each is a coarsening of the Oracle’s signal and no fit of $\mathcal{F}$ to it can do better asymptotically. The second half is the familiar approximation-estimation split. Distillation does not add information; the teacher is trained on $\mathcal{M}$, which the Oracle also sees. It substitutes a better *estimator* of η , computed in a class the student could not afford to serve, and hands the student a denoised regression target in place of binary labels.

This is the asymmetry that motivates offline teachers. Proposition 1’s ceiling is set by F , a property of the advertisers and the corpus. Proposition 2’s ceiling is set by $\mathcal{G}$, a property of the training budget. Only the second is something a practitioner can purchase, and the industrial trend towards large centrally trained models [5] is precisely a supply of it.

Both propositions are asymptotic and ignore optimisation error; Prop. 1 additionally assumes (3) and attributes the whole penalty to effective sample size. Section 5 shows where each idealisation is visible in the data.

4 Experimental Design

Protocol. We use rolling origins $T = 40, \dots, 59$, each with training windows ending at T minus that strategy’s own wait, and evaluation on the single day $[(T+1)d, (T+2)d)$ with one idle day in between, held fixed as the window slides. The label (1) is enforced identically in training and evaluation: a row may enter a training set only if its label is knowable at T . Evaluation days carry 230k to 300k rows. Conversions are logged to day 90.955 while clicks stop at day 61.001, so every evaluation label is fully resolved at every Δ . The origin is the unit of replication and all comparisons are paired on it, because the evaluation positive rate swings from 0.1595 to 0.1966 across these twenty days, an order of magnitude more than any effect we measure.

Tasks. We use $\Delta \in \{1d, 7d\}$ and drop $\Delta = 1h$, whose Oracle gap is -0.00002 ($p = 0.842$): there is no penalty to recover, so every strategy ties with Wait by construction.

Capacity Ladder. Strategy rankings that hold at one model size need not hold at another, so the entire comparison is repeated at five capacities $S_0, \dots, S_4$.

All five share one backbone and differ only in the number of categorical hash-table rows (36, 64, 160, 896 and 131,072, or 2,592 to 9,437,184 table parameters, a $3,641\times$ range), a single monotone knob that is exactly the embedding-table memory footprint and therefore the binding constraint in production ranking. The ladder was constructed to be approximately equally spaced in normalised entropy (mean step 0.0149, from 0.78626 at S_0 to 0.72794 at S_4 at one reference origin) and is monotone in it at 40 of 40 (task, origin) cells.

Teacher. Distil always uses the same teacher, selected as the best of a 21-rung architecture ladder and held at full table resolution: 57.3M parameters and 1.59M FLOPs per example, against 704 FLOPs for every student, a ratio of $2,262\times$. It is a training-only asset in exactly the sense Sect. 1 describes, and it could not be served under the student’s budget. The teacher sees Wait’s window and Wait’s labels, so it differs from Wait on capacity alone, receiving no extra data and no extra label information. We define its *headroom* as $H = \text{NE}(\text{Wait}) - \text{NE}(\text{teacher})$, the quantity Sect. 5.4 shows governs Distil.

Metrics. We report normalised entropy $\text{NE} = \text{logloss}/H(\bar{y})$, the calibration ratio $\bar{p}/\bar{y}$, and a calibration-adjusted NE obtained by multiplying every prediction by $c = \bar{y}/\bar{p}$, so that calibration is exactly one, and recomputing NE. That quantity can exceed NE, because matching the mean is not the same as minimising log loss; it is a level-matching diagnostic, not a lower bound. All models train for exactly one epoch: a second epoch already overfits at every capacity (train -0.0186 , evaluation $+0.0019$, $p = 0.025$).

Scale and Verification. The grid is 2 tasks $\times$ 5 capacities $\times$ 20 origins $\times$ 8 arms, or 1,600 scored runs. Four gates were checked before launch: the calibration multiplier forces calibration to one at every starting bias; the harness reproduces previously logged cells bit-exactly (14 of 14); the shared teacher is identical across capacities to seven decimal places while students are not; and every arm at a given (task, origin) is scored on an identical evaluation set.

5 Results

5.1 The Penalty Depends on the Window, Not Much on Capacity

The Oracle-minus-Wait gap at $\Delta = 7$ d is $+0.00538$, $+0.00602$, $+0.00746$, $+0.00842$ and $+0.00598$ for S_0 to S_4 , all with $p < 0.001$ over 20 of 20 origins. At $\Delta = 1$ d it is an order of magnitude smaller, from $+0.00067$ to $+0.00092$, though equally consistent (20 of 20, $p < 0.001$). Waiting is expensive only when the window is long relative to the available history: at seven days it withholds 12.6 to 18.6% of the corpus. Because the $\Delta = 1$ d denominator is below 0.001 NE, we report absolute differences at that window and treat recovery *ratios* there as indicative only.

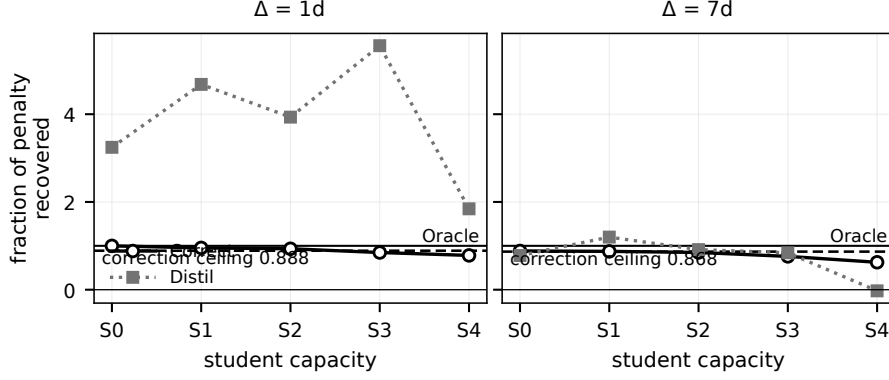


Fig. 1. The two ceilings. Correct (open circles) tracks the parameter-free prediction of Proposition 1 (dashed) at the three smallest capacities and never crosses the Oracle line at 1.0. Distil (grey squares) is not governed by either bound: it exceeds the Oracle throughout $\Delta=1d$, and collapses at S_4 for $\Delta=7d$, where the teacher has almost no headroom left

5.2 Correction Attains Its Ceiling, and Then Falls Below It

For $\Delta = 7d$ the predicted ceiling is 0.868, and Correct measures 0.883, 0.874, 0.849, 0.759 and 0.625 across S_0 through S_4 . At the three smallest capacities the prediction is accurate to $+0.015$, $+0.006$ and -0.019 , from a formula with no free parameters that was fitted to nothing. At S_3 and S_4 it over-predicts by 0.11 and 0.24.

The shortfall is misspecification, not estimation. Proposition 1 assumes (3), and that assumption is false on this corpus in a way we can measure. Stratifying converted rows by the distinct values of a single categorical field gives six strata holding 23,790 to 860,851 conversions each, across which the share of within-seven-day conversions landing in the first hour ranges from 0.453 to 0.689 (pooled: 0.574) and the median delay from 0.16 h to 1.73 h, a tenfold spread. A model too small to resolve covariate-dependent delay cannot be harmed by a correction that assumes it away, so at S_0 through S_2 the only cost of censoring is the statistical one that Proposition 1 prices exactly. As capacity grows, the misspecification bias in $\hat{F}(e)$ becomes expressible and recovery falls below the ceiling. The prediction is therefore best read as an upper bound that is tight precisely when the model is sample-limited rather than bias-limited, and the gap between prediction and measurement estimates what a covariate-conditional delay model would buy.

The practical reading is the same either way: where the ceiling is attained nothing is left for a better correction to win, and where it is not, the remedy is a covariate-conditional delay model rather than a tighter estimator. Both roads end at a bound set by F .

5.3 Distillation Is Not Bounded by the Oracle

No correction-based strategy exceeds the Oracle at $\Delta = 7$ d: Correct is below it at every capacity by 0.00063 to 0.00224 ($p < 0.005$), Band by 0.0035 to 0.0082, and Reweight by 0.0039 to 0.0081, all $p < 0.001$. This is Proposition 2’s first half, and Correct’s placement just under the bound is consistent with it being an efficient rather than a wasteful estimator.

Distil behaves differently in both windows. At $\Delta = 1$ d it beats the infeasible Oracle at S_0 through S_3 by +0.00150, +0.00294, +0.00269 and +0.00397, all $p < 0.001$, and by +0.00071 ($p = 0.06$) at S_4 . At $\Delta = 7$ d it is statistically indistinguishable from the Oracle at S_0 through S_3 ($p = 0.45$ to 0.73) while every other strategy is significantly below it. Against Wait it gains +0.00217 to +0.00484 at $\Delta = 1$ d and +0.00419 to +0.00722 at $\Delta = 7$ d over S_0 through S_3 , all $p < 0.001$ at 19 or 20 of 20 origins.

Head to head with Correct, Distil wins outright at the short window and ties at the long one: it is ahead by +0.00150 to +0.00411 at S_0 through S_3 for $\Delta = 1$ d ($p < 0.001$, 20 of 20 origins) and by +0.00089 at S_4 ($p = 0.019$), but the two are within ± 0.0020 with no significant difference at $\Delta = 7$ d ($p = 0.25$ to 0.80). The asymmetry follows from Sect. 5.2: at seven days Correct is near a ceiling worth 0.868 of a large penalty, whereas at one day the penalty is small and the teacher’s advantage is not.

One result cuts against Proposition 2 and we report it rather than reconcile it. At $\Delta = 1$ d, Reweight also exceeds the Oracle at S_0 (+0.00199) and S_2 (+0.00041), $p < 0.001$. The proposition is asymptotic, and Reweight trains on strictly less data under non-uniform weights, which regularises; at a window whose entire penalty is under 0.001 NE we read these as finite-sample effects. They do not recur at $\Delta = 7$ d, where Reweight is below the Oracle at all five capacities.

5.4 The Gain Is Bought With Teacher Headroom

Teacher headroom H falls monotonically as the student grows, from +0.0572 at S_0 to +0.0036 at S_4 for $\Delta = 7$ d, because the teacher is fixed while the student closes on it. Distil’s gain does not follow H down. Across S_1 to S_3 , where H falls by $2.8\times$ from 0.0484 to 0.0172, the gain is essentially flat at +0.00722, +0.00682 and +0.00709: a larger student extracts a larger fraction of a smaller advantage, with transfer efficiency gain/ H rising from 14.9% to 41.2% (and from 7.6% to 31.0% at $\Delta = 1$ d).

The flatness ends abruptly. At S_4 , $\Delta = 7$ d, headroom is +0.0036 and Distil collapses to -0.00015 ($p = 0.898$) — the only non-positive cell in the grid, and the only one where the Distil-Stale control is itself negative (-0.00126), meaning the teacher is actively degrading the target. Distillation helps exactly while the teacher is meaningfully better than the student. That is a statement about model size, not about feedback delay, and it is the boundary condition that makes teacher scale the operative variable: our teacher, at $2,262\times$ the student’s inference cost, is already spent against the largest student we tested.

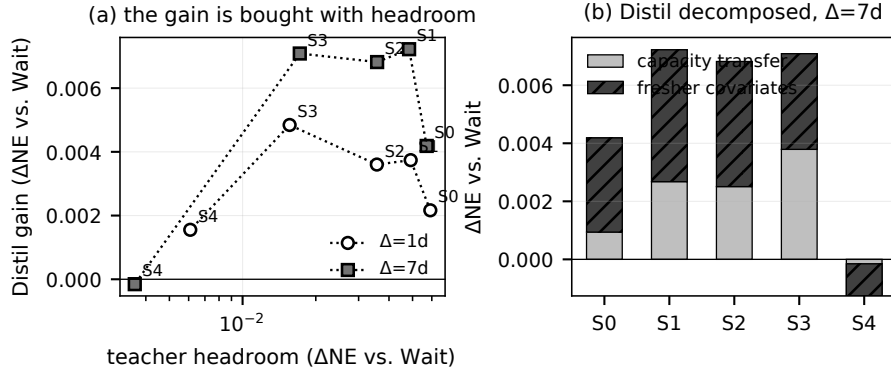


Fig. 2. (a) Distil’s gain against teacher headroom H (log axis), one point per capacity and window. Gain is roughly flat while H falls by $2.8\times$, then collapses at the single cell where H reaches $+0.0036$. (b) Distil decomposed with the Distil-Stale control at $\Delta=7d$. The fresher-covariate term is 47 to 78 % of the gain

5.5 What Distillation Is Actually Buying

Read alone, Distil’s win over Wait looks like delay mitigation. The Distil-Stale control shows it is only partly that. Capacity transfer accounts for 22 to 53 % of the gain at $\Delta = 7d$ across S_0 through S_3 , and 71 to 94 % at $\Delta = 1d$ (Fig. 2b). The residual, the student’s exposure to covariates that Wait discarded and which carry no new label information, is what is attributable to the delayed label; it is significant at every capacity ($p \leq 0.0013$, 19 or 20 of 20 origins) and worth $+0.0033$ to $+0.0045$ at $\Delta = 7d$.

We stress this because the decomposition is not visible without the control, and its two halves point in the same practical direction. The delay-attributable half is real and is the largest such term in our grid. The capacity half is the part that grows with the teacher. Reporting Distil against Wait alone would overstate delay mitigation by a factor of two to four at $\Delta = 1d$; we would encourage this literature to treat a Distil-Stale-style control as mandatory.

5.6 Band and Reweight Do Not Repay Their Complexity

Band ranges from -0.00215 to $+0.00457$ with no monotone pattern in capacity and is significant in only 5 of 10 cells; its slowest band still trains on Wait’s window, so it never escapes the wait it was designed to avoid. Reweight stays within -0.00055 to $+0.00265$ of Wait everywhere, consistent with the penalty being missing labels rather than covariate staleness — and Reweight by construction never touches an unresolved label.

6 Related Work

[To be written. Anchor on the joint conversion-delay likelihood [1]; the importance-weighting line [4, 8, 6]; stream-level alternatives [3, 2]; covariate-conditional delay [7, 9], which targets exactly the assumption Sect. 5.2 shows is violated; and the large centrally trained industrial ads models [5] that motivate Sect. 5.4. Distillation references are still to be added, and verified.]

7 Limitations and Outlook

The central extrapolation of this paper is untested. We show that Distil’s gain is governed by teacher headroom and that our teacher’s headroom is exhausted at S_4 ; we do not show what an LLM-scale teacher would deliver, because we did not run one. What the grid establishes is that the mechanism is real, that it is not bounded by the quantity that bounds every alternative here, and that the alternative is already at its own ceiling. Whether the trend continues at three more orders of magnitude of teacher is the experiment this baseline is meant to justify, not one it performs.

On the measurements: we use one seed per origin, with the origin as the unit of replication ($n = 20$); seed sensitivity was characterised separately at a standard deviation of 0.0004 to 0.0030, an order of magnitude below the $\Delta = 7$ d effects, but seeds are not averaged here, and the twenty origins share training data heavily, so the effective sample size is below twenty. Proposition 1 assumes a $1/n$ excess-risk rate that we do not verify directly, its empirical accuracy at S_0 through S_2 being the evidence offered; Proposition 2 is asymptotic, and Sect. 5.3 reports a finite-sample exception to it. The calibration multiplier is fitted on evaluation labels and is a diagnostic, never an achievable score. Distil’s soft targets are largely teacher in-sample on $\mathcal{M}$, and out-of-fold targets, the standard remedy, are untested. Finally, all results come from a single public corpus with one delay distribution, so the ceiling of Proposition 1 is corpus-specific even though the formula is not.

Acknowledgments. *[To be added after review.]*

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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